Lesson 1 — Extra Practice

Like Terms, and Adding/Subtracting Polynomials

Part A — Like Terms When There Is More Than One Variable

Two terms are **like** only when *every* variable carries *exactly* the same exponent. In $5x^2y$ and $5xy^2$ the letters match but the exponents do not, so these are **unlike** terms and cannot be collected.

1. Sort these terms into groups of like terms, then state how many groups there are.
$4x^2y, \quad -3xy, \quad x^2y, \quad 7xy^2, \quad 2xy, \quad -xy^2, \quad 5x^2, \quad -8x^2y$

2. Collect like terms. Watch the exponents carefully — not everything that looks similar can be combined.
(a) $5x^2y - 3xy^2 + 2x^2y + xy^2$ (b) $6a^2b + 4ab^2 - a^2b - 9ab^2 + 2ab$

(c) $3mn - 5m + 2n - 8mn + m - 7n$ (d) $-4x^2 + 7xy - y^2 + 9x^2 - 7xy + 3y^2$

3. For each polynomial, state the degree of **each term**, the degree of the **whole polynomial**, and whether it is a monomial, binomial, trinomial, or simply a polynomial. Remember that the degree of a term is the *sum* of its exponents.
(a) $5x^2y^3 - 2x^4 + 7$ (b) $-3a^2b^2 + ab - 6a^3$

(c) $8m^5 - m^2n^4$ (d) $x^3y - 7xy + 4y^2 - x$

4. State whether each pair is a pair of like terms. If not, explain exactly what differs.
(a) $3x^2y$ and $-7x^2y$ (b) $5ab$ and $5ba$ (c) $4x^2y$ and $4xy^2$ (d) $-6mn$ and $2mn^2$

Part B — Three Polynomials, Nested Brackets, and the Minus Sign

5. Simplify. Each expression contains **three** polynomials, so work left to right and change the signs of every bracket that follows a minus.

(a) $(4x^2 - 3x + 7) + (2x^2 + 5x - 1) - (x^2 - 2x + 9)$

(b) $(6y^2 + y - 8) - (3y^2 - 4y + 2) - (y^2 + 5y - 10)$

(c) $(x^2 + 2x) - (3x^2 - x + 5) + (2x^2 - 3x - 4)$

6. Simplify. Remove the **inner** brackets first, collect what you can, and only then deal with the outer bracket.

(a) $(3x^2 - x + 4) - \big[(x^2 + 2x) - (5 - x)\big]$

(b) $5a - [2a - (3a - 7)] + 4$

(c) $\big[(2x^2 + 5x) - (x^2 - 3)\big] - \big[(4x - 6) + (x^2 + x)\big]$

7. Simplify. In each part a minus sign sits in front of a bracket — including, in some parts, the *very first* bracket.

(a) $-(4x^2 - 7x + 2) + (x^2 - 3x)$

(b) $(2m^2 - 5m) - (-3m^2 + m - 8)$

(c) $-(x^2 + 6x - 4) - (3x^2 - 6x + 4)$

(d) $8 - (5x^2 - 2x + 8)$

8. Simplify each expression, then classify the **answer** by its degree and by its number of terms. Do not classify before you simplify — the degree can drop.

(a) $(3x^4 + 2x^2 - x) - (3x^4 - x^2 + 5)$

(b) $(x^3 + 4x^2) - (x^3 + 4x^2 - 7x)$

(c) $(2x^2y + 3xy) + (5xy - 2x^2y)$

(d) $(6x^5 - x^3 + 2) - (6x^5 - x^3)$

Part C — Working Backwards and Finding the Error

9. A student simplified $(7x^2 - 2x + 9) - (3x^2 - 5x + 4)$ like this:

$$7x^2 - 2x + 9 - 3x^2 - 5x + 4 \;=\; 4x^2 - 7x + 13.$$

(a) Which terms of the second bracket did the student fail to change?

(b) Rewrite the subtraction correctly and give the correct simplified answer.

(c) Substitute $x = 1$ into the original expression and into the student's answer to show they are not equal.

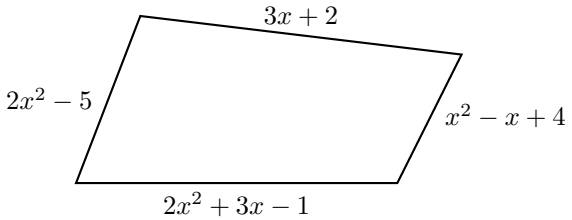
10. Find the missing polynomial that makes each statement true. Check your answer by substituting it back in.
(a) $(2x^2 + 5x - 3) + \square = 5x^2 - x$ (b) $\square - (x^2 - 4x + 7) = 2x^2 + x - 1$ (c) $(4a^2 - a + 6) - \square = a^2 + 3a$

11. (a) What must be **added to** $3x^2 - 8x + 2$ to give $x^2 + 4$?
(b) What must be **subtracted from** $6y^2 + y - 5$ to give $2y^2 - 3y$?
(c) In part (b), explain why the answer is *not* simply the difference taken the other way around.

12. **Stretch.** Determine the missing coefficients.
(a) Find a and b so that $(ax^2 + 3x) - (2x^2 + bx) = x^2 - 5x$.
(b) Find a, b and c so that $(5x^2 + ax - 7) + (bx^2 - 2x + c) = 8x^2 + x - 3$.

Part D — Applications, Checking, and Reasoning

13. The quadrilateral below has its four side lengths given in centimetres.



(a) Write and simplify an expression for the perimeter.

(b) Determine the perimeter when $x = 3$.

14. A rectangle has a length of $(3x + 7)$ m and a width of $(x - 2)$ m.

(a) By adding all four sides, write and simplify an expression for the perimeter.

(b) Determine the perimeter when $x = 5$, and explain why $x = 1$ would not make sense here.

15. **Substitution check.** Simplifying does not change the value of an expression, so the original and the simplified form must agree for *every* value of the variable.

(a) Simplify $(5x^2 - 3x + 8) - (2x^2 + 4x - 1)$.

(b) Evaluate the **original** expression at $x = -2$ by working out each bracket separately, then subtracting.

(c) Evaluate your **simplified** answer at $x = -2$. Do the two values agree?

(d) If the two values had disagreed, what would that tell you — and what would it *not* prove if they had agreed only at $x = 0$?

16. Subtract using the **vertical method**. Notice that the first polynomial has no x^2 term — leave a gap in that column, or write $0x^2$.

$$\begin{array}{r} 6x^3 - 4x + 1 \\ -\,(2x^3 + 5x^2 - 1) \\ \hline \end{array}$$

17. **Reasoning.** Answer each question in a full sentence, with an example where one is asked for.

(a) Can the sum of two trinomials be a binomial? If so, give an example; if not, explain why not.

(b) Can the sum of two trinomials be a monomial? If so, give an example; if not, explain why not.

(c) A polynomial of degree 4 is added to a polynomial of degree 6. What is the degree of the sum? Explain how you know.

(d) Two polynomials of degree 5 are added. What is the **highest possible** degree of the sum? Give an example where the degree of the sum is lower than that.

Answer Key. 1. Four groups — x^2y: $4x^2y$, x^2y, $-8x^2y$; xy^2: $7xy^2$, $-xy^2$; xy: $-3xy$, $2xy$; x^2: $5x^2$ (on its own). **2.** (a) $7x^2y - 2xy^2$ (b) $5a^2b - 5ab^2 + 2ab$ (c) $-5mn - 4m - 5n$ (d) $5x^2 + 2y^2$ (the xy terms cancel). **3.** (a) degrees 5, 4, 0; polynomial degree 5; trinomial. (b) degrees 4, 2, 3; degree 4; trinomial. (c) degrees 5, 6; degree 6; binomial. (d) degrees 4, 2, 2, 1; degree 4; polynomial (4 terms). **4.** (a) like (b) like — ab and ba are the same product (c) not like — x has exponent 2 in one and 1 in the other (d) not like — n has exponent 1 in one and 2 in the other. **5.** (a) $5x^2 + 4x - 3$ (b) $2y^2$ (the y terms and the constants both cancel) (c) -9 (everything with a variable cancels). **6.** (a) inner bracket $= x^2 + 3x - 5$, so the answer is $2x^2 - 4x + 9$ (b) inner gives $-a + 7$, so $5a + a - 7 + 4 = 6a - 3$ (c) first bracket $= x^2 + 5x + 3$, second $= x^2 + 5x - 6$, so the answer is 9. **7.** (a) $-3x^2 + 4x - 2$ (b) $5m^2 - 6m + 8$ (c) $-4x^2$ (d) $-5x^2 + 2x$. **8.** (a) $3x^2 - x - 5$: degree 2, trinomial (the quartic terms cancel) (b) $7x$: degree 1, monomial (c) $8xy$: degree 2, monomial (d) 2: degree 0, monomial (a constant). **9.** (a) Only the $3x^2$ had its sign changed; the $-5x$ and the $+4$ were left alone. (b) $-(3x^2 - 5x + 4) = -3x^2 + 5x - 4$, so $7x^2 - 2x + 9 - 3x^2 + 5x - 4 = 4x^2 + 3x + 5$. (c) At $x = 1$ the original is $14 - 2 = 12$ and the correct answer gives 12, but the student's answer gives 10. **10.** (a) $3x^2 - 6x + 3$ (b) $3x^2 - 3x + 6$ (c) $3a^2 - 4a + 6$. **11.** (a) $-2x^2 + 8x + 2$ (b) $4y^2 + 4y - 5$ (c) Subtraction is not reversible: the missing polynomial is what is taken *away*, so it equals (first) − (result), not (result) − (first); reversing it gives $-4y^2 - 4y + 5$, the opposite. **12.** (a) $a - 2 = 1$ so $a = 3$; $3 - b = -5$ so $b = 8$. (b) $5 + b = 8$ so $b = 3$; $a - 2 = 1$ so $a = 3$; $-7 + c = -3$ so $c = 4$. **13.** (a) $(2x^2 + 3x - 1) + (x^2 - x + 4) + (3x + 2) + (2x^2 - 5) = 5x^2 + 5x$ cm (b) $45 + 15 = 60$ cm. **14.** (a) $(3x + 7) + (x - 2) + (3x + 7) + (x - 2) = 8x + 10$ m (b) 50 m; at $x = 1$ the width would be $1 - 2 = -1$ m, and a length cannot be negative. **15.** (a) $3x^2 - 7x + 9$ (b) at $x = -2$: $5(4) + 6 + 8 = 34$ and $2(4) - 8 - 1 = -1$, so $34 - (-1) = 35$ (c) $3(4) + 14 + 9 = 35$ — the two values agree (d) A disagreement would mean an error was made in simplifying (most likely a sign). Agreeing at a single value such as $x = 0$ proves nothing, since two different expressions can happen to match at one point; it is only a check, not a proof. **16.** $4x^3 - 5x^2 - 4x + 2$. **17.** (a) Yes — one pair of like terms can cancel, e.g. $(x^2 + 3x + 5) + (2x^2 - 3x + 1) = 3x^2 + 6$. (b) Yes — two pairs can cancel, e.g. $(x^2 + 3x + 5) + (2x^2 - 3x - 5) = 3x^2$. (c) Degree 6: the degree-6 term has nothing of the same degree to cancel with, so it survives. (d) At most degree 5; it can be lower if the degree-5 terms cancel, e.g. $(2x^5 + 3x^2 - 1) + (-2x^5 + x + 4) = 3x^2 + x + 3$, which has degree 2.

Lesson 2 — Extra Practice

Multiplying a Monomial by a Polynomial

Part A — Negative Monomials and Two Variables

1. Expand. Watch every sign — a negative monomial flips the sign of *each* term inside the brackets.
(a) $-3x^2(2x^2 - 4x + 5)$ (b) $-5x(x^2 - 3x - 2)$ (c) $-x^3(4x^2 + x - 7)$

2. Expand. Two variables change nothing about the method: multiply the coefficients, then add the exponents on x and on y *separately*.
(a) $4xy(3x^2 - 2xy + y^2)$ (b) $-2ab(5a^2 - 3ab + 4b^2)$ (c) $3m^2n(2mn - n^2 + 6)$

3. Expand. Each monomial has a fractional coefficient, but the rule is unchanged: multiply the fraction by every coefficient inside.
(a) $\frac{1}{2}x(6x^2 - 4x + 10)$ (b) $-\frac{2}{3}a^2(-6a^2 + 9a + 3)$ (c) $\frac{1}{4}x(8x^2 + 12x - 4)$

Part B — Distribute, Then Collect

Two brackets means two separate distributions. Expand the first one completely and write it down before you touch the second — then, and only then, collect like terms. For example,

$$3x(x-2) + 4x(2x+1) \;=\; \underbrace{3x^2 - 6x}_{\text{first}} + \underbrace{8x^2 + 4x}_{\text{second}} \;=\; 11x^2 - 2x.$$

4. Expand and simplify.
(a) $2x(3x + 1) + 5x(x - 4)$ (b) $3a(a - 2) - 4a(2a + 1)$ (c) $-4y(2y - 3) - 3y(y + 5)$

5. Expand and simplify. These produce cubic terms, so keep your like terms lined up by degree.
(a) $5x(x^2 - 2x) + 3x^2(x + 4)$ (b) $-2x(x^2 + 3x - 1) - x(4x^2 - 5x)$ (c) $6t(2t - 3) - t(t - 8)$

6. **Nested brackets.** Simplify *inside* the square brackets first, then distribute the monomial across what is left.
(a) $2x\,[3x - (x + 4)]$ (b) $-3a\,[2a - (4a - 5)]$ (c) $5x\,[x + 2(x - 3)]$

7. Simplify inside the square brackets first, then distribute.
(a) $4x\,[2(x+3)-3(x-1)]$ (b) $-2y\,[y-3(y+2)]$

Part C — Finding Errors, Working Backwards, and Checking

8. Each student below made exactly one mistake. Describe the mistake in a sentence, then give the correct expansion.
(a) Priya wrote $4x(x^2-3x+2)=4x^3-3x+2$. (b) Devon wrote $-2x(3x-5)=-6x^2-10x$.

9. **Work backwards.** Find the monomial that belongs in each box.
(a) $\square(2x-5)=6x^2-15x$ (b) $3x(\square)=12x^3-9x^2$ (c) $-2x^2(\square)=-8x^4+6x^3-10x^2$

10. Find the missing factor. In each case, ask yourself what the first term of the bracket had to be multiplied by.
(a) $\square(x+3)=-5x^2-15x$ (b) $4ab(\square)=12a^2b-8ab^2$

11. **Substitution check.** Expand $3x(x^2-2x+4)$. Then evaluate *both* the original expression and your expanded expression at $x=3$. If the two values disagree, your expansion is wrong — find and fix the error.

12. Expand $-2x(x^2+3x-1)$, then check your answer by evaluating both the original and the expanded form at $x=-2$. State both values.

Part D — Applications, Reasoning, and Stretch

13. A rectangle has width $2x$ and length $(3x+4)$.
(a) Write a simplified expression for its area.
(b) Write a simplified expression for its perimeter.
(c) Determine the area when $x=5$.

14. A large rectangle measures $5x$ by $(4x+3)$. A smaller rectangle measuring $2x$ by $(x+1)$ is cut out of it. Write a simplified expression for the area of the **remaining** region, then find that area when $x=4$.

15. **Reasoning.** A monomial of degree 2 is multiplied by a trinomial of degree 3.
(a) What is the degree of the product? Explain how you know without expanding anything.

(b) How many terms will the product have, before any collecting? Explain.

16. **Reasoning.** Nia claims that a monomial times a trinomial *always* gives a trinomial. Jae claims the answer could end up with fewer than three terms once like terms are collected. Decide who is right and justify your answer.

17. **Stretch.**
(a) Expand $-2x^2y(3xy^2 - 4x^2y + y^3)$.

(b) Expand and simplify $x(x+1)(x+2)$ by distributing **twice**: multiply the first two factors together, then multiply that result by the third. *This is a preview of Lesson 3.*

Answer Key. 1. (a) $-6x^4+12x^3-15x^2$ (b) $-5x^3+15x^2+10x$ (c) $-4x^5-x^4+7x^3$. **2.** (a) $12x^3y-8x^2y^2+4xy^3$ (b) $-10a^3b+6a^2b^2-8ab^3$ (c) $6m^3n^2-3m^2n^3+18m^2n$. **3.** (a) $3x^3-2x^2+5x$ (b) $4a^4-6a^3-2a^2$ (c) $2x^3+3x^2-x$. **4.** (a) $6x^2+2x+5x^2-20x=11x^2-18x$ (b) $3a^2-6a-8a^2-4a=-5a^2-10a$ (c) $-8y^2+12y-3y^2-15y=-11y^2-3y$. **5.** (a) $5x^3-10x^2+3x^3+12x^2=8x^3+2x^2$ (b) $-2x^3-6x^2+2x-4x^3+5x^2=-6x^3-x^2+2x$ (c) $12t^2-18t-t^2+8t=11t^2-10t$. **6.** (a) inside: $2x-4$, so $4x^2-8x$ (b) inside: $-2a+5$, so $6a^2-15a$ (c) inside: $3x-6$, so $15x^2-30x$. **7.** (a) inside: $2x+6-3x+3=-x+9$, so $-4x^2+36x$ (b) inside: $y-3y-6=-2y-6$, so $4y^2+12y$. **8.** (a) She multiplied only the first term of the bracket by $4x$ and copied the other two down unchanged; every term must be multiplied: $4x^3-12x^2+8x$. (b) He kept the second sign negative, but $(-2x)(-5)=+10x$: $-6x^2+10x$. **9.** (a) $3x$ (b) $4x^2-3x$ (c) $4x^2-3x+5$. **10.** (a) $-5x$ (b) $3a-2b$. **11.** $3x^3-6x^2+12x$. At $x=3$: original $=9(9-6+4)=9(7)=63$; expanded $=81-54+36=63$. They agree. **12.** $-2x^3-6x^2+2x$. At $x=-2$: original $=4(4-6-1)=4(-3)=-12$; expanded $=16-24-4=-12$. They agree. **13.** (a) $A=2x(3x+4)=6x^2+8x$ (b) $P=2(2x)+2(3x+4)=10x+8$ (c) $6(25)+8(5)=190$. **14.** $5x(4x+3)-2x(x+1)=20x^2+15x-2x^2-2x=18x^2+13x$; at $x=4$, $18(16)+13(4)=340$. **15.** (a) Degree 5. Multiplying adds exponents, so the highest-degree term of the trinomial (degree 3) times the monomial (degree 2) gives degree $3+2=5$. (b) Three terms — the distributive property produces one product for each term in the brackets. **16.** Nia is right. The three terms of a trinomial have different variable parts, and multiplying all three by the *same* monomial keeps them different, so no two products can ever be like terms and there is nothing to collect. (Jae would be right only if the monomial were 0, or if the "trinomial" had not been simplified first.) **17.** (a) $-6x^3y^3+8x^4y^2-2x^2y^4$, usually written $8x^4y^2-6x^3y^3-2x^2y^4$. (b) First $x(x+1)=x^2+x$; then $(x^2+x)(x+2)=x^3+2x^2+x^2+2x=x^3+3x^2+2x$.

Lesson 3 — Extra Practice

Multiplying Two Binomials

Part A — Heavier Coefficients and Two Variables

1. Expand and simplify. Every one of these has a coefficient in front of x in *both* binomials, so all four FOIL products need care.

(a) $(3x-4)(2x+5)$ (b) $(5x-2)(4x-3)$ (c) $(4x+7)(3x-2)$

(d) $(6x-5)(2x+9)$ (e) $(7x+3)(3x-8)$ (f) $(8x-1)(5x-6)$

2. Expand and simplify. Treat the second variable exactly like a number you cannot combine with x — the middle terms are xy terms and the last term carries y^2.

(a) $(2x+3y)(x-4y)$ (b) $(5a-2b)(3a+b)$ (c) $(4m+3n)(2m-5n)$

3. Expand and simplify.

(a) $(3p-7q)(3p+7q)$ (b) **Stretch.** $(2x^2+5)(x-3)$ (c) **Stretch.** $(x+4)(x^2-2)$

Part B — A Monomial in Front

4. Each expression is a monomial multiplied by *two* binomials. Expand the two binomials first, then distribute the monomial through the trinomial you get.

(a) $3x(x+2)(x-5)$ (b) $-2x(x-3)(x+4)$

(c) $4y(2y-1)(y+3)$ (d) $-5(2x-1)(x-6)$

5. Ravi expanded $3x(x+2)(x-5)$ by first multiplying $3x$ by $(x+2)$ to get $3x^2+6x$, and then multiplying that by $(x-5)$. Ali expanded the two binomials first and then distributed the $3x$. Do they get the same answer? Explain why the order does not matter here.

6. Expand $2x(x+1)(x+3)$ and $2(x+1)(x+3)$. State in one sentence exactly what the extra factor of x does to each term of the answer.

Part C — Two Products at Once

7. Expand *both* products completely before you collect like terms. A subtraction sign in front of a product flips the sign of all four of its terms.

(a) $(x+3)(x+4)-(x+2)(x+1)$

(b) $(x+9)(x-2)-(x+5)(x-6)$

(c) $(2x+1)(x+5)-(x+3)(x+2)$

(d) $(3x-2)(x+4)+(x-1)(x+7)$

8. Expand and simplify.

(a) $(4x-3)(2x+1)-(3x+2)(x-5)$

(b) $2(x+5)(x-3)-3(x+1)(x-2)$

9. In question 7(a) the x^2 terms vanished completely. Expand $(x+7)(x+1)-(x+4)(x+2)$ and show that the same thing happens. Then explain, using the First products, why subtracting any two products of the form $(x+a)(x+b)-(x+c)(x+d)$ always removes the x^2 term.

Part D — Checking, Errors, and a Shortcut Worth Noticing

10. **Error analysis.** A student writes $(x+3)(x+4)=x^2+12$. This is the single most common mistake in this course.

(a) Which two of the four FOIL products were left out?

(b) Give the correct expansion.

(c) Substitute $x=1$ into the student's answer and into yours to show they are different.

11. Identify the error in each expansion and give the correct result.

(a) $(2x-5)(3x-4)=6x^2-23x-20$

(b) $(x+6)(x-6)=x^2-12x-36$

(c) $(3x+2)(x+5)=3x^2+17x+7$

12. **Substitution check.** Expand $(3x-4)(2x+5)$. Then evaluate the *original* product and your *expanded* answer, both at $x=2$, and confirm that the two values match. Explain why a matching value is strong evidence but not a proof.

13. Use a substitution check at $x = 3$ to decide whether each expansion is correct. Show both values. If it is wrong, expand it properly.
(a) $(x-1)(x+8) = x^2 + 7x - 8$ (b) $(2x+3)(x-4) = 2x^2 - 5x + 12$

14. **A shortcut to come.** Expand each product the long way with FOIL.
(a) $(x+5)(x+5)$ (b) $(x-7)(x-7)$ (c) $(2x+3)(2x+3)$

Look at your three answers beside the binomials that produced them. In each case, how is the x^2 term related to the first term of the binomial? the last number related to the second term? the middle term related to *both*? Write the pattern as a rule for $(a+b)(a+b)$. *This is the shortcut you will meet in Lesson 4 — you have just discovered it.*

Part E — Working Backwards

15. Find the numbers a and b in each case. *Hint:* the last number is ab and the middle coefficient is $a+b$ — look for a factor pair of the last number that adds to the middle coefficient.
(a) $(x+a)(x+b) = x^2 + 9x + 20$ (b) $(x+a)(x+b) = x^2 + 2x - 15$ (c) $(x+a)(x+b) = x^2 - 11x + 24$

16. Determine the missing binomial in each product.
(a) $(2x+3)(\square) = 6x^2 + x - 12$ (b) $(\square)(x-5) = 4x^2 - 17x - 15$ (c) $(x+7)(\square) = x^2 + 4x - 21$

17. Determine the value of k so that the expansion of $(x+k)(x-6)$ has **no x-term**. Expand the product using your value of k and state the two-term answer.

18. Determine the value of c so that $(2x+c)(x+5)$ expands to a trinomial whose constant term is -20. Then state the full expansion.

19. **Reasoning.** The distributive property always produces four products, yet the final answer rarely has four terms.

(a) When does the product of two binomials simplify to only **three** terms? Give an example.

(b) When does it collapse to only **two** terms? Give an example and describe the pattern (you'll meet its name in Lesson 4).

(c) Give an example of two binomials whose product genuinely keeps all **four** terms.

Part F — Area Applications and a Stretch

20. A rectangle has length $(2x - 1)$ cm and width $(x + 3)$ cm.

(a) Determine a simplified expression for its area.

(b) Determine a simplified expression for its perimeter.

(c) Evaluate both when $x = 4$.

21. A rectangular photo measures $(x + 2)$ cm by $(x + 5)$ cm. It is enlarged so that the length and the width each increase by 3 cm.

(a) Write and expand an expression for the new area.

(b) Determine a simplified expression for how much the area increased.

22. A rectangular garden measures $(x + 2)$ m by $(x + 5)$ m. A concrete path 1 m wide is poured all the way around the *outside* of the garden.

(a) Write expressions for the length and the width of the outer rectangle (garden plus path).

(b) Determine a simplified expression for the area of the path alone.

(c) The path costs \$35 per square metre. If $x = 6$, what is the total cost?

23. **Stretch.** Three binomials can be multiplied by expanding two of them first, then distributing the trinomial that results over the remaining binomial.

(a) Expand and simplify $(x + 1)(x + 2)(x + 3)$.

(b) Expand and simplify $(x - 2)(x + 4)(2x - 1)$.

24. **Stretch.** A student claims that some **positive** value of k makes $(x + k)(x + 6)$ expand with no x-term. Explain why no positive k can work, then find the one value of k that does and state the resulting two-term product. *Compare your answer with question 17.*

Answer Key. 1. (a) $6x^2+7x-20$ (b) $20x^2-23x+6$ (c) $12x^2+13x-14$ (d) $12x^2+44x-45$ (e) $21x^2-47x-24$ (f) $40x^2-53x+6$. **2.** (a) $2x^2-5xy-12y^2$ (b) $15a^2-ab-2b^2$ (c) $8m^2-14mn-15n^2$. **3.** (a) $9p^2-49q^2$ (b) $2x^3-6x^2+5x-15$ (c) x^3+4x^2-2x-8. **4.** (a) $3x^3-9x^2-30x$ (b) $-2x^3-2x^2+24x$ (c) $8y^3+20y^2-12y$ (d) $-10x^2+65x-30$. **5.** Yes — both give $3x^3-9x^2-30x$. Multiplication is associative and commutative, so the three factors may be grouped in any order. **6.** $2x(x+1)(x+3)=2x^3+8x^2+6x$ and $2(x+1)(x+3)=2x^2+8x+6$; the extra factor of x raises the degree of every term by one (each exponent goes up by 1). **7.** (a) $4x+10$ (b) $8x+12$ (c) x^2+6x-1 (d) $4x^2+16x-15$. **8.** (a) $5x^2+11x+7$ (b) $-x^2+7x-24$. **9.** $(x+7)(x+1)-(x+4)(x+2)=(x^2+8x+7)-(x^2+6x+8)=2x-1$. Each First product is $x\cdot x=x^2$, so the two expansions always start with the same x^2 term; subtracting removes it, leaving at most a linear expression. **10.** (a) the Outer ($4x$) and Inner ($3x$) products (b) $x^2+7x+12$ (c) at $x=1$ the student's answer gives $1+12=13$, but $(1+3)(1+4)=20$ and $1+7+12=20$ — not equal. **11.** (a) sign error on Last: $-5\times-4=+20$, so $6x^2-23x+20$ (b) O and I are $-6x$ and $+6x$, which cancel: x^2-36 (c) Last is $2\times5=10$, not 7: $3x^2+17x+10$. **12.** $(3x-4)(2x+5)=6x^2+7x-20$. At $x=2$: original $=(2)(9)=18$; expanded $=24+14-20=18$. Matching at one value only rules out that value — two different expressions can still agree at a single point, so a full expansion is what proves it. **13.** (a) Correct. At $x=3$: original $=(2)(11)=22$ and $9+21-8=22$. (b) Incorrect. At $x=3$: original $=(9)(-1)=-9$ but $18-15+12=15$. The Last product is $3\times(-4)=-12$, not $+12$, so the correct expansion is $2x^2-5x-12$. **14.** (a) $x^2+10x+25$ (b) $x^2-14x+49$ (c) $4x^2+12x+9$. The first term is the first term squared, the last term is the second term squared, and the middle term is twice their product: $(a+b)(a+b)=a^2+2ab+b^2$. **15.** (a) $a=4$, $b=5$ (b) $a=5$, $b=-3$ (c) $a=-3$, $b=-8$ (order interchangeable). **16.** (a) $3x-4$ (b) $4x+3$ (c) $x-3$. **17.** $k=6$: $(x+6)(x-6)=x^2-36$ (the $+6x$ and $-6x$ cancel). **18.** $c=-4$: $(2x-4)(x+5)=2x^2+6x-20$. **19.** (a) Three terms whenever the Outer and Inner products are like terms but do not cancel — e.g. $(x+3)(x+4)=x^2+7x+12$. (b) Two terms when O and I are opposites and cancel, which happens for $(a+b)(a-b)$ — e.g. $(x+6)(x-6)=x^2-36$, a difference of squares. (c) Four terms when O and I are not like terms — e.g. $(x+2)(y+3)=xy+3x+2y+6$. **20.** (a) $A=(2x-1)(x+3)=2x^2+5x-3$ cm^2 (b) $P=2(2x-1)+2(x+3)=6x+4$ cm (c) at $x=4$: $A=32+20-3=49$ cm^2 and $P=28$ cm. **21.** (a) New area $=(x+5)(x+8)=x^2+13x+40$ cm^2 (b) increase $=(x^2+13x+40)-(x^2+7x+10)=6x+30$ cm^2. **22.** (a) Outer rectangle is $(x+4)$ m by $(x+7)$ m. (b) Path $=(x+4)(x+7)-(x+2)(x+5)=(x^2+11x+28)-(x^2+7x+10)=4x+18$ m^2. (c) At $x=6$ the path is 42 m^2, so the cost is $42\times\$35=\$1470$. **23.** (a) $x^3+6x^2+11x+6$ (b) $2x^3+3x^2-18x+8$. **24.** $(x+k)(x+6)=x^2+(k+6)x+6k$. The x-coefficient is $k+6$, and if k is positive then $k+6>6$, so it can never be 0 — no positive k works. Setting $k+6=0$ gives $k=-6$, and then $(x-6)(x+6)=x^2-36$. A zero x-term needs the two constants to be *opposites*, which is exactly the arrangement in question 17.

Lesson 4 — Extra Practice

Special Products: Difference of Squares and Perfect Squares

Part A — Leading Coefficients and Two Variables

The patterns $(a+b)(a-b) = a^2 - b^2$ and $(a \pm b)^2 = a^2 \pm 2ab + b^2$ do not care what a and b are. In $(3x+5)(3x-5)$ the first term is $a = 3x$, so $a^2 = 9x^2$ — the coefficient gets squared too.

1. Expand using the difference of squares pattern.
(a) $(3x+5)(3x-5)$ (b) $(4y+9)(4y-9)$ (c) $(2x-11)(2x+11)$ (d) $(6a+7)(6a-7)$

2. Expand using the perfect square trinomial pattern.
(a) $(4x-7)^2$ (b) $(5x+6)^2$ (c) $(7x-2)^2$ (d) $(10x+3)^2$

3. Now both terms carry a variable. Square each term completely — including its letter.
(a) $(5a-2b)(5a+2b)$ (b) $(7p+3q)(7p-3q)$ (c) $(2x+3y)^2$ (d) $(3m+4n)^2$

4. The terms being squared can be anything at all. *Recall that* $(x^2)^2 = x^4$.
(a) $(6c-5d)^2$ (b) $(x^2+9)(x^2-9)$ (c) $(2x^2-5)^2$ (d) $(ab+10)(ab-10)$

Part B — Deciding Which Pattern Applies

5. For each product, first write which case it is — **difference of squares**, **perfect square**, or **neither** — then expand and simplify.

(a) $(4x+9)(4x-9)$
Pattern: _____

(b) $(5x-1)^2$
Pattern: _____

(c) $(x+7)(x-3)$
Pattern: _____

(d) $(3x-8)(3x+8)$
Pattern: _____

(e) $(6x+5)^2$
Pattern: _____

(f) $(2x-7)(2x+5)$
Pattern: _____

6. Look back at your six answers in question 5. Two of them have only **two** terms and four of them have **three**. Which pattern produces a two-term answer, and why does the middle term vanish for that one but not the others?

7. Without expanding, explain how you can tell at a glance that $(x+6)(x-6)$ and $(x-6)(x+6)$ must give the same answer, and that $(x+6)^2$ and $(x-6)^2$ must *not*.

Part C — The Missing Middle Term

The single most common error in this lesson is writing $(x+5)^2 = x^2 + 25$. Squaring a binomial is *not* the same as squaring each term, and a well-chosen number settles the argument instantly.

8. Substitute $x = 1$ into both expressions and compare.

(a) Evaluate $(x+5)^2$ at $x = 1$.

(b) Evaluate $x^2 + 25$ at $x = 1$.

(c) Expand $(x+5)^2$ correctly.

(d) Check your answer to (c) at $x = 1$.

9. Your correct expansion in 8(c) and the wrong version x^2+25 differ by exactly one term. Set the two expressions equal to each other and solve, to find the **only** value of x for which the wrong answer happens to be right.

10. **Error analysis.** A student wrote $(2x-5)^2 = 4x^2-25$. Name the mistake, give the correct expansion, and then substitute $x = 1$ into the student's answer and yours to show they disagree.

Part D — Special Products as Mental Arithmetic

Here is the pay-off. To find 49×51, notice that both numbers sit one step from 50:

$$49 \times 51 = (50-1)(50+1) = 50^2 - 1^2 = 2500 - 1 = 2499.$$

Squares work the same way: $21^2 = (20+1)^2 = 400 + 2(20)(1) + 1 = 441$. No calculator, no long multiplication.

11. Rewrite each product as $(a-b)(a+b)$ and evaluate it mentally. Show the rewriting.

(a) 38×42

(b) 97×103

(c) 195×205

(d) 999×1001

12. Rewrite each square as $(a \pm b)^2$ with a round a, then evaluate it mentally.

(a) 19^2

(b) 102^2

(c) 28^2

(d) 53^2

13. You know that $35^2 = 1225$. Use the difference of squares pattern to write down 34×36 in one step, and explain why the answer is exactly 1 less than 1225.

14. The trick in question 11 needs the two numbers to be the *same distance* from a round number. Explain why 34×36 can be done this way but 34×37 cannot, then state the round number and the distance for 88×92.

Part E — Combining Special Products

When two special products are added or subtracted, expand each one *fully* first, then collect like terms. Something surprising usually happens.

15. Expand each special product, then simplify.

(a) $(x+3)^2 - (x-3)^2$

(b) $(x+2)^2 + (x-2)^2$

16. Expand and simplify. Watch the signs when you subtract a whole trinomial.

(a) $(2x+1)^2 - (2x-1)^2$

(b) $(x+5)(x-5) + (x+5)^2$

17. Expand and simplify. In part (b), multiply through by the number in front *after* expanding the bracket.

(a) $(x+4)^2 - (x+1)(x-1)$

(b) $3(x-2)^2 - 2(x+3)(x-3)$

Part F — Working Backwards

18. Each expansion below is already finished. Determine what belongs in the box.

(a) $(\square + 7)(\square - 7)$ $= 4x^2 - 49$

(b) $(\square - 3)(\square + 3)$ $= 25x^2 - 9$

(c) $(\square + 6)^2$ $= 9x^2 + 36x + 36$

(d) $(x - \square)^2$ $= x^2 - 14x + 49$

19. A trinomial is a **perfect square** when it comes from squaring a binomial. Use that idea to find the missing value in each case.

(a) $x^2 + kx + 36$ is a perfect square trinomial. Find **both** values of k.

(b) $4x^2 + kx + 25$ is a perfect square trinomial. Find **both** values of k.

(c) $9x^2 + 30x + c$ is a perfect square trinomial. Find c, and name the binomial.

20. In question 19(a) there were two answers, but in 19(c) there was only one. Explain the difference between asking for the *middle* term and asking for the *last* term of a perfect square trinomial.

21. Explain why $x^2 + 9$ can **never** be written as $(x+a)(x-a)$ for any real number a, even though $x^2 - 9$ can. *Hint: what would a^2 have to equal?*

Part G — Area and Geometry

22. A square piece of cardboard has side length $(x + 4)$ cm. A small square of side 4 cm is cut out of one corner and thrown away.

(a) Write an expression for the area of the original square, expanded and simplified.

(b) Write and simplify an expression for the area that is left after the corner is removed.

23. A square garden has side length s metres. Each side is then increased by 3 m. Write and simplify an expression for **how much the area increases**, and state whether that increase is the same for every garden or depends on s.

24. Compare a square with side x to a rectangle measuring $(x + 5)$ by $(x - 5)$. Show that the square always has the larger area, and determine exactly how much larger — the answer does not contain x at all.

Part H — Stretch

25. The letters a and b in $(a+b)(a-b) = a^2-b^2$ can stand for *whole expressions*, not just single terms. In $(x+y+2)(x+y-2)$, treat the entire group $(x + y)$ as one object a, and 2 as b.

(a) Apply the pattern to get $(x + y)^2 - 2^2$, then expand $(x + y)^2$ and write the final simplified answer.

(b) Use the same idea on $(a + b + 3)(a + b - 3)$.

(c) Use the same idea on $\big((x + 1) + 4\big)\big((x + 1) - 4\big)$, treating $(x + 1)$ as the single object a. Then check your answer by expanding $(x + 5)(x - 3)$ directly — the two must agree.

Answer Key. 1. (a) $9x^2 - 25$ (b) $16y^2 - 81$ (c) $4x^2 - 121$ (d) $36a^2 - 49$. **2.** (a) $16x^2 - 56x + 49$ (b) $25x^2 + 60x + 36$ (c) $49x^2 - 28x + 4$ (d) $100x^2 + 60x + 9$. **3.** (a) $25a^2 - 4b^2$ (b) $49p^2 - 9q^2$ (c) $4x^2 + 12xy + 9y^2$ (d) $9m^2 + 24mn + 16n^2$. **4.** (a) $36c^2 - 60cd + 25d^2$ (b) $x^4 - 81$ (c) $4x^4 - 20x^2 + 25$ (d) $a^2b^2 - 100$. **5.** (a) difference of squares, $16x^2 - 81$ (b) perfect square, $25x^2 - 10x + 1$ (c) neither, $x^2 + 4x - 21$ (d) difference of squares, $9x^2 - 64$ (e) perfect square, $36x^2 + 60x + 25$ (f) neither, $4x^2 - 4x - 35$. **6.** The difference of squares, parts (a) and (d). The Outer and Inner products are opposites ($+ab$ and $-ab$), so they cancel; in the other cases they do not. **7.** Multiplication is commutative, so $(x+6)(x-6)$ and $(x-6)(x+6)$ are literally the same product. But $(x+6)^2$ and $(x-6)^2$ are different products: they give $x^2 + 12x + 36$ and $x^2 - 12x + 36$, which differ in the middle term. **8.** (a) $(1+5)^2 = 36$ (b) $1 + 25 = 26$ (c) $x^2 + 10x + 25$ (d) $1 + 10 + 25 = 36$, which matches (a), so the expansion is right and $x^2 + 25$ is wrong. **9.** $x^2 + 10x + 25 = x^2 + 25$ gives $10x = 0$, so $x = 0$ is the only value. **10.** The student squared each term and lost the middle term $2ab = -20x$; correctly $(2x - 5)^2 = 4x^2 - 20x + 25$. At $x = 1$: student's $4 - 25 = -21$, correct $4 - 20 + 25 = 9$, and $(2 - 5)^2 = (-3)^2 = 9$. The student is wrong. **11.** (a) $(40-2)(40+2) = 1600-4 = 1596$ (b) $(100-3)(100+3) = 10\,000-9 = 9991$ (c) $(200-5)(200+5) = 40\,000-25 = 39\,975$ (d) $(1000-1)(1000+1) = 1\,000\,000-1 = 999\,999$. **12.** (a) $(20-1)^2 = 400 - 40 + 1 = 361$ (b) $(100+2)^2 = 10\,000 + 400 + 4 = 10\,404$ (c) $(30-2)^2 = 900 - 120 + 4 = 784$ (d) $(50+3)^2 = 2500 + 300 + 9 = 2809$. **13.** $34 \times 36 = (35-1)(35+1) = 35^2 - 1^2 = 1225 - 1 = 1224$. The pattern subtracts $b^2 = 1^2 = 1$ from the square, so the product is always exactly 1 below 35^2. **14.** 34 and 36 are both 1 away from 35, so they fit $(a-b)(a+b)$ with $a = 35$, $b = 1$. 34 and 37 have no common centre — their midpoint 35.5 is not a whole number — so no whole a and b work. For 88×92: $a = 90$, $b = 2$ (and $8100 - 4 = 8096$). **15.** (a) $(x^2 + 6x + 9) - (x^2 - 6x + 9) = 12x$ (b) $(x^2 + 4x + 4) + (x^2 - 4x + 4) = 2x^2 + 8$. **16.** (a) $(4x^2 + 4x + 1) - (4x^2 - 4x + 1) = 8x$ (b) $(x^2 - 25) + (x^2 + 10x + 25) = 2x^2 + 10x$. **17.** (a) $(x^2 + 8x + 16) - (x^2 - 1) = 8x + 17$ (b) $3(x^2 - 4x + 4) - 2(x^2 - 9) = 3x^2 - 12x + 12 - 2x^2 + 18 = x^2 - 12x + 30$. **18.** (a) $2x$ (or $-2x$) (b) $5x$ (or $-5x$) (c) $3x$ (d) 7. **19.** (a) $k = 12$ or $k = -12$ (from $(x \pm 6)^2$) (b) $k = 20$ or $k = -20$ (from $(2x \pm 5)^2$) (c) $c = 25$, and $9x^2 + 30x + 25 = (3x + 5)^2$ (or $(-3x - 5)^2$). **20.** The middle term is $2ab$, and b may be positive or negative, so both + and − work — two answers. The last term is b^2, and squaring either sign of b gives the same positive number — one answer. **21.** $(x + a)(x - a) = x^2 - a^2$, so we would need $-a^2 = +9$, i.e. $a^2 = -9$. No real number squares to a negative, so it is impossible. For $x^2 - 9$ we need $a^2 = 9$, giving $a = 3$. **22.** (a) $(x + 4)^2 = x^2 + 8x + 16$ cm^2 (b) $x^2 + 8x + 16 - 16 = x^2 + 8x$ cm^2. **23.** $(s + 3)^2 - s^2 = (s^2 + 6s + 9) - s^2 = 6s + 9$ m^2. It depends on s: a bigger garden gains more area from the same 3 m increase. **24.** Square $= x^2$; rectangle $= (x + 5)(x - 5) = x^2 - 25$. The difference is $x^2 - (x^2 - 25) = 25$, so the square is always 25 square units larger, no matter what x is. **25.** (a) $(x + y)^2 - 4 = x^2 + 2xy + y^2 - 4$ (b) $(a + b)^2 - 9 = a^2 + 2ab + b^2 - 9$ (c) $(x + 1)^2 - 16 = x^2 + 2x + 1 - 16 = x^2 + 2x - 15$, and $(x + 5)(x - 3) = x^2 + 2x - 15$ — they agree.

Lesson 5 — Extra Practice

Multiplying a Binomial by a Trinomial

Part A — Leading Coefficients in Both Brackets

Every question in this part has a coefficient in front of the x in the binomial *and* in front of the x^2 in the trinomial. There are still exactly six partial products — they are just harder to multiply. Expand and simplify, writing each answer in **descending order of degree**.

1. (a) $(2x-3)\left(3x^2+x-4\right)$ (b) $(4x+1)\left(2x^2-3x+5\right)$

2. (a) $(3x-2)\left(2x^2-5x-1\right)$ (b) $(5x+2)\left(3x^2+x-2\right)$

3. (a) $(2x+5)\left(4x^2-3x+2\right)$ (b) $(6x-1)\left(x^2-2x-3\right)$

4. **Before you multiply.** Answer (a) and (b) *without* expanding anything, then do the expansion in (c).

(a) How many partial products does a binomial times a trinomial produce before any like terms are collected? Explain how you know from the number of terms in each bracket.

(b) What is the degree of the simplified answer, and at most how many terms will it have once like terms are collected?

(c) Now expand and simplify $(3x+4)\left(2x^2-x+5\right)$, listing all six products before you collect. State the coefficient of x^2 in your final answer.

Part B — Negative Signs Throughout

Now the trinomial itself can start with a negative term. Do not try to guess the sign pattern of the answer — determine the sign of **each** of the six partial products separately, then collect.

5. (a) $(x-2)\left(-x^2+4x-5\right)$ (b) $(-2x+1)\left(x^2-3x-4\right)$

6. (a) $(-x-3)\left(-x^2-2x+6\right)$ (b) $(2x-7)\left(-3x^2+x-1\right)$

7. **Error analysis.** Jasleen expanded $(x+5)\left(x^2+2x+3\right)$ as shown. Her answer is wrong.

$(x+5)\left(x^2+2x+3\right)$
$= x^3+2x^2+3x+5x^2+15$ Line 1
$= x^3+7x^2+3x+15$ Line 2

(a) How many partial products should Line 1 contain, and how many does it contain?
(b) Name the product she left out.
(c) Write the corrected expansion.

8. **Error analysis.** Owen expanded $(x-3)\left(x^2-4x+2\right)$ as shown.

$(x-3)\left(x^2-4x+2\right)$
$= x^3-4x^2+2x+3x^2-12x+6$ Line 1
$= x^3-x^2-10x+6$ Line 2

(a) Which three of Owen's six partial products are wrong, and what single mistake caused all three?
(b) Write the corrected expansion.

Part C — Two Variables: A Result Worth Remembering

Nothing changes when a second variable appears — $x \cdot y = xy$, and xy terms collect with other xy terms only. Expand and simplify questions 9–11 completely.

9. (a) $(x+y)\left(x^2-xy+y^2\right)$ (b) $(x-y)\left(x^2+xy+y^2\right)$

(c) Both answers in question 9 are remarkably short. Describe what happened to the four middle terms in each one, and write down the two patterns you have just proved. (They are called the **sum** and **difference of cubes**, and you will meet them again when you learn to factor.)

10. Each product below is a disguised version of one of the patterns from question 9. Expand and simplify, then say which pattern it matched.
(a) $(a+2b)\left(a^2-2ab+4b^2\right)$ (b) $(2m-n)\left(4m^2+2mn+n^2\right)$

11. These two look similar to question 10 but do *not* match either pattern. Expand and simplify, then explain in one sentence why the middle terms survive this time.
(a) $(x+3y)\left(x^2-2xy+5y^2\right)$ (b) $(2p-q)\left(p^2+3pq-q^2\right)$

Part D — Monomials in Front, and Cubing a Binomial

12. A monomial factor in front makes this a **three**-factor product. Multiply the two brackets first, then distribute the monomial over all four terms. Notice that the degree climbs to 4.
(a) $2x(x+1)\left(x^2-3x+2\right)$ (b) $-3x(x-2)\left(x^2+x-1\right)$

13. **Cubing a binomial.** $(x+1)^3$ means $(x+1)(x+1)(x+1)$. Square the binomial first using the special product from Lesson 4, which turns the problem into a binomial times a trinomial. For example, $(x+1)^3=(x+1)\left(x^2+2x+1\right)$. Show both stages.
(a) $(x+2)^3$ (b) $(x-3)^3$

14. Cube each binomial using the same two-stage method. Take care with the coefficients when you square.
(a) $(2x-1)^3$ (b) $(3x+2)^3$

15. **Checking by substitution.** An expansion is an identity: the original expression and the simplified answer must give the same value for *every* number you substitute.

(a) Expand and simplify $(x-4)\left(x^2+2x-3\right)$.

(b) Evaluate the *original* expression at $x=2$ by working out each bracket first.

(c) Evaluate your *expanded* answer at $x=2$. Do the two values agree?

(d) A classmate says a substitution check that works at $x=2$ proves the expansion is correct. Explain why one successful substitution is good evidence but not a proof, and suggest what would make the check more convincing.

Part E — Working Backwards, Volume, and a Stretch

16. **Find the missing bracket.** In each case a binomial and a trinomial were multiplied to give the polynomial shown. Determine the missing bracket, then check by expanding. *Hint:* start with the highest-degree term and the constant term, since each comes from only one partial product.

(a) $(x+3)\,(_____) = x^3+2x^2-9x-18$

(b) $(_____)\left(x^2-x+4\right) = 2x^3-5x^2+11x-12$

17. **Volume.** A rectangular box has height x centimetres, width $(x+2)$ centimetres, and length $(x+3)$ centimetres.

(a) Multiply the width and the length to get a trinomial for the area of the base.

(b) Multiply that trinomial by the height to get a cubic polynomial for the volume, in descending order.

(c) Determine the volume when $x=4$ cm using your cubic, then check it by multiplying the three actual dimensions.

(d) Explain why x cannot be 0 or a negative number in this situation.

18. **Targeting one coefficient.** Sometimes only a single coefficient is needed, and expanding everything is wasted work.

(a) In the expansion of $(2x-5)\left(3x^2+4x-1\right)$, determine the coefficient of x^2 and the constant term *without* writing out all six products. Explain which partial products you had to use for each.

(b) In the expansion of $(x+k)\left(x^2+3x-2\right)$, the coefficient of x is 10. Determine the value of k, then write the full expansion to confirm it.

19. **Stretch.** Each part below pushes the same distributive idea past three terms.
(a) Expand and simplify $(x+1)\left(x^3-x^2+x-1\right)$. How many partial products are there this time? The simplified answer has only two terms — explain what cancelled.

(b) Show that $(a+b)^3 = a^3+3a^2b+3ab^2+b^3$ by first squaring $(a+b)$ and then multiplying the resulting trinomial by $(a+b)$. Use your result to write down $(a-b)^3$ without doing the multiplication again, and explain your reasoning.

Answer Key. 1. (a) $6x^3-7x^2-11x+12$ (b) $8x^3-10x^2+17x+5$. **2.** (a) $6x^3-19x^2+7x+2$ (b) $15x^3+11x^2-8x-4$. **3.** (a) $8x^3+14x^2-11x+10$ (b) $6x^3-13x^2-16x+3$. **4.** (a) Six — each of the 2 binomial terms multiplies each of the 3 trinomial terms, and $2\times 3=6$. (b) Degree 3 (from $x\cdot x^2$); at most four terms, since the six products collect into one x^3, one x^2, one x and one constant. (c) $6x^3-3x^2+15x+8x^2-4x+20 = 6x^3+5x^2+11x+20$; the coefficient of x^2 is 5. **5.** (a) $-x^3+6x^2-13x+10$ (b) $-2x^3+7x^2+5x-4$. **6.** (a) x^3+5x^2-18 (the x terms are $-6x+6x=0$, so there is no x term) (b) $-6x^3+23x^2-9x+7$. **7.** (a) Six are required; only five appear. (b) She left out $5\cdot 2x=10x$. (c) $x^3+2x^2+3x+5x^2+10x+15 = x^3+7x^2+13x+15$. **8.** (a) All three products from the second term are wrong: he used $+3$ instead of -3, so $3x^2$, $-12x$ and $+6$ should be $-3x^2$, $+12x$ and -6. (b) $x^3-4x^2+2x-3x^2+12x-6 = x^3-7x^2+14x-6$. **9.** (a) x^3+y^3 (b) x^3-y^3. (c) In each expansion the two x^2y terms are opposites and cancel, and so do the two xy^2 terms, leaving only the first and last products. The patterns are $(x+y)\left(x^2-xy+y^2\right)=x^3+y^3$ and $(x-y)\left(x^2+xy+y^2\right)=x^3-y^3$. **10.** (a) a^3+8b^3 — sum of cubes with $y=2b$, since $(2b)^3=8b^3$. (b) $8m^3-n^3$ — difference of cubes with $x=2m$. **11.** (a) $x^3+x^2y-xy^2+15y^3$ (b) $2p^3+5p^2q-5pq^2+q^3$. The middle terms survive because the trinomial is not the exact partner of the binomial: in the true patterns the last term of the trinomial is the square of the binomial's second term and the middle term is $\mp$ the product of the two, which is what makes the middle terms opposites. **12.** (a) $(x+1)\left(x^2-3x+2\right)=x^3-2x^2-x+2$, so the answer is $2x^4-4x^3-2x^2+4x$. (b) $(x-2)\left(x^2+x-1\right)=x^3-x^2-3x+2$, so the answer is $-3x^4+3x^3+9x^2-6x$. **13.** (a) $(x+2)\left(x^2+4x+4\right)=x^3+6x^2+12x+8$ (b) $(x-3)\left(x^2-6x+9\right)=x^3-9x^2+27x-27$. **14.** (a) $(2x-1)\left(4x^2-4x+1\right)=8x^3-12x^2+6x-1$ (b) $(3x+2)\left(9x^2+12x+4\right)=27x^3+54x^2+36x+8$. **15.** (a) $x^3-2x^2-11x+12$. (b) $(2-4)(4+4-3)=(-2)(5)=-10$. (c) $8-8-22+12=-10$; yes, they agree. (d) A wrong expansion can still happen to give the right value at one particular number, so a single check only rules out some errors. Testing a second and third value — $x=0$ and $x=1$ are quick, and $x=-1$ catches sign errors — makes it far more convincing; a full re-expansion is the actual proof. **16.** (a) x^2-x-6, since $x\cdot x^2=x^3$ forces the leading x^2 and $3\cdot(-6)=-18$ forces the constant; check: $(x+3)\left(x^2-x-6\right)=x^3+2x^2-9x-18$. (b) $2x-3$, since $2x\cdot x^2=2x^3$ and $(-3)(4)=-12$; check: $(2x-3)\left(x^2-x+4\right)=2x^3-5x^2+11x-12$. **17.** (a) $(x+2)(x+3)=x^2+5x+6$. (b) $V=x\left(x^2+5x+6\right)=x^3+5x^2+6x$ cubic centimetres. (c) $64+80+24=168$; and $4\times 6\times 7=168$ cm^3. (d) A length must be positive, so $x>0$; at $x=0$ the box would have no height and no volume, and a negative x has no meaning as a measurement. **18.** (a) Coefficient of x^2: only $2x\cdot 4x=8x^2$ and $-5\cdot 3x^2=-15x^2$ produce x^2, giving $8-15=-7$. Constant: only $-5\cdot(-1)=5$ has no x, so the constant is 5. (Full expansion, for checking: $6x^3-7x^2-22x+5$.) (b) The x terms are $x\cdot(-2)=-2x$ and $k\cdot 3x=3kx$, so $3k-2=10$ and $k=4$; then $(x+4)\left(x^2+3x-2\right)=x^3+7x^2+10x-8$, which indeed has $10x$. **19.** (a) Eight partial products (2×4); the answer is x^4-1. Writing them out gives $x^4-x^3+x^2-x+x^3-x^2+x-1$, in which the x^3 terms cancel, the x^2 terms cancel and the x terms cancel, leaving only x^4 and -1. (b) $(a+b)^2=a^2+2ab+b^2$, and $(a+b)\left(a^2+2ab+b^2\right)=a^3+2a^2b+ab^2+a^2b+2ab^2+b^3=a^3+3a^2b+3ab^2+b^3$. Replacing b with $-b$ flips the sign of every term containing an odd power of b, so $(a-b)^3=a^3-3a^2b+3ab^2-b^3$.

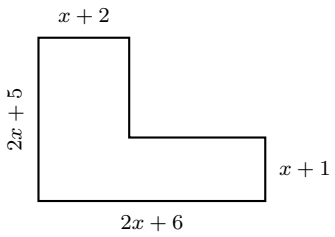
Lesson 6 — Extra Practice

Applying Polynomial Products to Area, Volume, and Equations

Part A — Composite Areas: Adding and Subtracting Regions

When a region is not a plain rectangle, cut it into rectangles you can handle. Add the pieces if the region is built *up* from rectangles; subtract if a piece has been removed. Any side length that is not labelled can usually be found by subtracting two that are.

1. The L-shaped patio below is made of two rectangles. Only four sides are labelled.

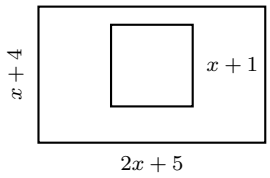


(a) Find the length of the unlabelled vertical side (the height of the upper rectangle), in terms of x.

(b) Write a simplified expression for the total area of the patio.

(c) Determine the area when $x = 3$ m.

2. A rectangular sheet of metal has a square hole punched out of the middle, as shown.



(a) Write a simplified expression for the area of metal that remains.

(b) Determine the remaining area when $x = 4$ cm.

3. A photograph measures 12 cm by 8 cm. A wooden frame of uniform width x cm is built around all four sides of it.
(a) Write expressions for the outer length and the outer width of the framed picture.
(b) Write a simplified expression for the total outer area.
(c) Write a simplified expression for the area of the frame alone (the wood only).
(d) Test $x = 1, 2, 3, \ldots$ in your expression from (c) to find the width that uses 96 cm^2.

4. A rectangular swimming pool measures $(x + 5)$ m by $(x + 2)$ m. A concrete deck of uniform width 3 m is poured all the way around the pool. (a) Write expressions for the outer dimensions of the pool-and-deck together. (b) Write a simplified expression for the area of the deck alone. (c) Notice that your answer to (b) is *linear*. Explain why the x^2 terms disappeared. (d) Determine the deck area when $x = 4$.

5. A large square of side $(x+6)$ cm has a smaller square of side $(x-2)$ cm removed from one corner. (a) Write a simplified expression for the shaded area that remains. (b) Now redo part (a) using the difference of squares pattern $a^2-b^2=(a+b)(a-b)$ instead of expanding, and confirm you get the same answer. (c) Determine the shaded area when $x=5$.

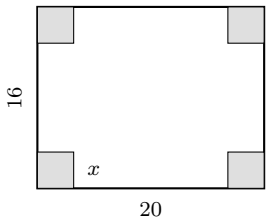
6. **Error analysis.** A rectangular tabletop measures $(x+7)$ by $(x+5)$. A square tile of side $(x+3)$ is glued to the centre of it. Dana found the uncovered area like this:

$$(x+7)(x+5)-(x+3)^2 \;=\; x^2+12x+35-x^2+6x+9 \;=\; 18x+44.$$

(a) Identify Dana's error and name the step where it happened. (b) Give the correct simplified expression. (c) Evaluate the correct expression when $x=2$.

Part B — Volume and Surface Area

7. **The open-top box.** A rectangular sheet of cardboard measures 20 cm by 16 cm. A square of side x cm is cut from each corner, and the four flaps are folded up to form an open-top box.



(a) Write expressions for the length, the width, and the height of the finished box.

(b) Write the volume of the box as a simplified polynomial in x.

(c) Determine the volume when $x=3$ cm.

(d) Explain why x must be less than 8 cm for a box to exist at all.

8. A square sheet of tin measures 24 in by 24 in. A square of side x in is cut from each corner and the flaps are folded up to make an open-top box. (a) Write the volume as a simplified polynomial in x. (b) Determine the volume when $x=4$ in. (c) State the possible values of x that give a real box.

9. A rectangular prism has length $(2x+1)$ cm, width $(x+3)$ cm, and height $(x+2)$ cm. (a) Write the volume as a simplified polynomial. (b) Determine the volume when $x = 2$ cm. (c) Check your answer to (b) by substituting $x = 2$ into the three dimensions first and multiplying.

10. **Surface area.** A closed rectangular box has length $(x + 4)$ cm, width $(x + 2)$ cm, and height $(x + 1)$ cm. The surface area of a closed box is $SA = 2\ell w + 2\ell h + 2wh$.
(a) Write and simplify an expression for the surface area of the closed box.
(b) The lid is now removed. Write and simplify an expression for the surface area of the open-top box (five faces).
(c) Evaluate both expressions when $x = 3$, and confirm that the difference between them is the area of the missing lid.

11. **Stretch.** A cube has edge length x cm. Every edge is then increased by 3 cm. (a) Write the volume of the enlarged cube as a simplified polynomial. *Hint:* expand $(x+3)^2$ first, then multiply by $(x+3)$ again. (b) Write a simplified expression for the *increase* in volume. (c) Determine the increase when $x = 5$ cm, and verify it directly by computing $8^3 - 5^3$.

12. **Stretch.** A cylinder has radius $(x + 2)$ cm and height 5 cm, so its volume is $V = \pi r^2 h$. (a) Write its volume as π times a simplified trinomial. (b) The radius is increased by 1 cm while the height stays at 5 cm. Write the new volume the same way. (c) Show that the increase in volume simplifies to $\pi(10x + 25)$ cm^3, and state what happens to the x^2 terms.

Part C — Solving Equations by Expanding First

Each equation below *looks* quadratic, but once both sides are expanded the x^2 terms are identical and cancel, leaving a linear equation. Expand first, then collect.

13. Solve each equation. Show the expansion, then verify your solution by substituting it back into the original equation.
(a) $(x + 3)(x + 4) = x^2 + 19$ (b) $(x - 4)(x + 6) = x^2 - 14$

14. Solve each equation. Show the expansion and verify your solution.
(a) $(x+2)^2-(x-1)^2=15$ (b) $(x+4)^2-(x+1)^2=27$

15. Solve each equation.
(a) $(x+5)(x-2)=(x+1)(x+3)$ (b) $(2x+3)(x+5)=2x^2+41$ (c) $(x+6)(x-6)=x^2-4x-8$

16. A square garden has side length $(x+5)$ m. A rectangular garden measures $(x+2)$ m by $(x+9)$ m. The two gardens have exactly the same area. (a) Write an equation and solve it for x. (b) State the actual dimensions of each garden. (c) State the common area, and confirm it is the same for both.

17. **Error analysis.** Kai solved $(x+6)(x+2)=x^2+20$ as follows:

$$x^2+8x+12=x^2+20 \quad \Rightarrow \quad 8x=32 \quad \Rightarrow \quad x=4.$$

(a) The expansion is correct, so the mistake is later. Identify it. (b) Solve the equation correctly. (c) Substitute both $x=4$ and your answer into the original equation to show which one actually works.

18. **Error analysis.** A rectangle has length $(x+7)$ cm and width $(x+3)$ cm. Miguel wrote: "The area is $2(x+7)+2(x+3)=4x+20$." (a) Explain what Miguel actually calculated and what the area should be. (b) Give the correct simplified area expression. (c) Determine the value of x for which the *perimeter* is 40 cm, then find the area of that rectangle.

Part D — Comparing, Reversing, and Proving

19. Two rectangles are built from nearly the same numbers: Rectangle P measures $(x+5)$ by $(x+3)$, and Square Q has side $(x+4)$.
(a) Expand the area of each.
(b) Which one is larger, and by how much? Notice that your answer does not depend on x at all.
(c) Test your conclusion with $x = 10$ and again with $x = 1$ by computing both areas numerically.
(d) Repeat the comparison for $(x+9)(x+7)$ and $(x+8)^2$. State the general result in a sentence.

20. **Working backwards.** Each expression below is a known product. Find the missing factor by asking "what must multiply the given side to produce this?"
(a) A rectangle has area $2x^2+11x+12$ and width $(x+4)$. Find its length, then write a simplified expression for its perimeter.
(b) A rectangular prism has volume $x^3+6x^2+11x+6$ and a base of area x^2+3x+2. Find its height.
(c) Check part (b) by multiplying your height back by the base area.

21. **Consecutive integers.** Let n be any whole number, so n and $n+1$ are consecutive integers. (a) Expand $n(n+1)$. (b) Explain why $n(n+1)$ is always even, no matter which integer n you choose. *Hint:* one of n and $n+1$ must be even. (c) Expand and simplify $(n+1)^2 - n^2$, and explain why the result is always odd.

22. **Stretch.** Two consecutive even integers can be written as $2n$ and $2n+2$. (a) Expand and simplify their product. (b) Show that $4n^2+4n = 4(n^2+n)$ by expanding the right side, so the product is always a multiple of 4. (c) Using question 21(b), explain why the product is in fact always a multiple of 8. (d) Test your claim with the pair 6 and 8, and again with 10 and 12.

23. **Stretch.** Let n be any number. (a) Expand $(n-2)(n+2)$. (b) Use your answer to explain, in one sentence, why the product of any two numbers that differ by 4 is always exactly 4 less than the square of the number halfway between them. (c) Test the claim with 18 and 22. (d) Predict, without multiplying, the value of 48×52.

Answer Key. 1. (a) The left edge is $2x+5$ and the right edge is $x+1$, so that side is $(2x+5)-(x+1)=x+4$. (b) $(2x+6)(x+1)+(x+2)(x+4)=2x^2+8x+6+x^2+6x+8=3x^2+14x+14$. (c) 83 m^2. **2.** (a) $(2x+5)(x+4)-(x+1)^2=2x^2+13x+20-(x^2+2x+1)=x^2+11x+19$. (b) 79 cm^2. **3.** (a) length $12+2x$, width $8+2x$. (b) $4x^2+40x+96$ cm^2. (c) $(4x^2+40x+96)-96=4x^2+40x$ cm^2. (d) $x=1$: $4+40=44$; $x=2$: $4(4)+80=96$ ✓, so $x=2$ cm. **4.** (a) $(x+11)$ m by $(x+8)$ m. (b) $(x+11)(x+8)-(x+5)(x+2)=x^2+19x+88-(x^2+7x+10)=12x+78$ m^2. (c) Both areas contain exactly one x^2, so subtracting removes it; the deck has constant width, so its area grows only in proportion to the perimeter. (d) 126 m^2. **5.** (a) $(x+6)^2-(x-2)^2=(x^2+12x+36)-(x^2-4x+4)=16x+32$ cm^2. (b) $[(x+6)+(x-2)][(x+6)-(x-2)]=(2x+4)(8)=16x+32$ — same answer. (c) 112 cm^2. **6.** (a) She subtracted only the first term of $(x+3)^2$: the minus sign must be distributed across all three terms of x^2+6x+9. (b) $x^2+12x+35-x^2-6x-9=6x+26$. (c) 38 square units. **7.** (a) length $20-2x$, width $16-2x$, height x. (b) $V=x(20-2x)(16-2x)=x(4x^2-72x+320)=4x^3-72x^2+320x$ cm^3. (c) 420 cm^3 (the box is $14\times10\times3$). (d) The width $16-2x$ must be positive, so $x<8$; otherwise there is no material left to fold up. **8.** (a) $V=x(24-2x)^2=4x^3-96x^2+576x$ in^3. (b) 1024 in^3 (the box is $16\times16\times4$). (c) $0<x<12$. **9.** (a) $(2x+1)(x+3)(x+2)=(2x+1)(x^2+5x+6)=2x^3+11x^2+17x+6$ cm^3. (b) 100 cm^3. (c) $5\times5\times4=100$ ✓ **10.** (a) $\ell w=x^2+6x+8$, $\ell h=x^2+5x+4$, $wh=x^2+3x+2$, so $SA=6x^2+28x+28$ cm^2. (b) Open top removes one ℓw face: $SA=5x^2+22x+20$ cm^2. (c) 166 cm^2 and 131 cm^2; the difference is 35 cm^2, which is $\ell w=7\times5$ ✓ **11.** (a) $(x+3)^3=(x^2+6x+9)(x+3)=x^3+9x^2+27x+27$ cm^3. (b) $9x^2+27x+27$ cm^3. (c) 387 cm^3; $8^3-5^3=512-125=387$ ✓ **12.** (a) $5\pi(x+2)^2=\pi(5x^2+20x+20)$ cm^3. (b) $5\pi(x+3)^2=\pi(5x^2+30x+45)$ cm^3. (c) The difference is $\pi(5x^2+30x+45)-\pi(5x^2+20x+20)=\pi(10x+25)$; the $5x^2$ terms are identical and cancel, so the increase is linear. **13.** (a) $x^2+7x+12=x^2+19$, so $7x=7$ and $x=1$; check $(4)(5)=20=1+19$ ✓ (b) $x^2+2x-24=x^2-14$, so $2x=10$ and $x=5$; check $(1)(11)=11=25-14$ ✓ **14.** (a) $(x^2+4x+4)-(x^2-2x+1)=6x+3=15$, so $x=2$; check $16-1=15$ ✓ (b) $(x^2+8x+16)-(x^2+2x+1)=6x+15=27$, so $x=2$; check $36-9=27$ ✓ **15.** (a) $x^2+3x-10=x^2+4x+3\Rightarrow-x=13\Rightarrow x=-13$. (b) $2x^2+13x+15=2x^2+41\Rightarrow13x=26\Rightarrow x=2$. (c) $x^2-36=x^2-4x-8\Rightarrow4x=28\Rightarrow x=7$. **16.** (a) $(x+5)^2=(x+2)(x+9)$ gives $x^2+10x+25=x^2+11x+18$, so $x=7$. (b) Square: 12 m by 12 m; rectangle: 9 m by 16 m. (c) Both have area 144 m^2 ✓ **17.** (a) He subtracted 12 from the wrong side: from $8x+12=20$ the next line is $8x=8$, not $8x=32$. (b) $x=1$. (c) $x=4$ gives $(10)(6)=60$ but $16+20=36$ — not equal; $x=1$ gives $(7)(3)=21$ and $1+20=21$ ✓ **18.** (a) He calculated the *perimeter* formula, not the area; area is length times width. (b) $(x+7)(x+3)=x^2+10x+21$ cm^2. (c) $4x+20=40\Rightarrow x=5$; the rectangle is 12 cm by 8 cm, so the area is 96 cm^2. **19.** (a) P: $x^2+8x+15$; Q: $x^2+8x+16$. (b) The square Q is larger, by exactly 1 square unit for every value of x. (c) $x=10$: $15\times13=195$ and $14^2=196$; $x=1$: $6\times4=24$ and $5^2=25$ ✓ (d) $x^2+16x+63$ versus $x^2+16x+64$ — again a difference of 1. In general $(x+a)^2$ is always exactly 1 more than $(x+a+1)(x+a-1)$. **20.** (a) Length $=2x+3$, since $(x+4)(2x+3)=2x^2+11x+12$; perimeter $=2(2x+3)+2(x+4)=6x+14$. (b) Height $=x+3$. (c) $(x^2+3x+2)(x+3)=x^3+6x^2+11x+6$ ✓ **21.** (a) n^2+n. (b) Of any two consecutive integers exactly one is even, and an even factor makes the whole product even — so $n(n+1)$ is always even. (c) $(n^2+2n+1)-n^2=2n+1$, which is one more than the even number $2n$, so it is always odd. **22.** (a) $2n(2n+2)=4n^2+4n$. (b) $4(n^2+n)=4n^2+4n$ ✓, so the product is 4 times the integer $n^2+n=n(n+1)$. (c) By 21(b), $n(n+1)$ is itself even, say $n(n+1)=2k$; then the product is $4(2k)=8k$, a multiple of 8. (d) $6\times8=48=8\times6$ ✓ and $10\times12=120=8\times15$ ✓ **23.** (a) n^2-4. (b) The two numbers are $n-2$ and $n+2$, which differ by 4 and have n halfway between them; their product n^2-4 is 4 less than n^2. (c) $18\times22=396$ and $20^2-4=400-4=396$ ✓ (d) $48\times52=50^2-4=2496$.

Lesson 7 — Extra Practice

Unit Review: Polynomial Operations

Every question in this set pulls together **two or more** skills from Unit 3. Expect to distribute, FOIL, use a pattern, and then collect like terms — all in the same question. Watch the subtraction signs: they are where most of the marks are lost.

Part A — Multi-Step Simplification

1. Expand each product, then collect like terms. Work one product at a time and keep the brackets until you have distributed the sign in front of them.

(a) $3x(x-2)+(x+4)(x-1)-(x-3)^2$

(b) $5x(x+3)-(2x-1)(x+6)+4\left(x^2-x\right)$

2. Expand and simplify. One part of each question uses a special-product pattern.

(a) $(3x+2)(x-5)-(x+4)^2$

(b) $(2x-7)(2x+7)-(x-3)(x+8)$

3. Each expression is a product of **three** factors. Multiply two of them first — choosing the pair that forms a special product makes the work much shorter — then multiply the result by the remaining factor.

(a) $2(x+3)(x-3)(x+1)$

(b) $-x(2x-1)(2x+1)$

4. A cube of a binomial is just a square followed by one more multiplication.

(a) Expand $(x+2)^2$.

(b) Multiply your answer from (a) by $(x+2)$ to obtain $(x+2)^3$.

(c) Use the same two-step method to expand $(x-1)^3$.

5. Expand and simplify. Notice that the first term is a binomial times a trinomial and the second is a monomial times a perfect square.

$$(2x+1)\left(x^2-3x+4\right)-x(x-5)^2$$

Part B — Special Products and Patterns

6. Expand using the appropriate pattern. Each of these has two different variables or a squared base, so be careful about what plays the role of a and what plays the role of b.

(a) $(3x + 2y)^2$ (b) $(5a - 4b)(5a + 4b)$ (c) $\left(x^2 + 3\right)\left(x^2 - 3\right)$

7. Work backwards from the pattern to find the missing terms.

(a) $(x + ____)^2 = x^2 + 14x + 49$ (b) $(3x - ____)^2 = 9x^2 - 24x + 16$ (c) $(____ + 5)\,(____ - 5) = 16x^2 - 25$

8. Patterns are not only for algebra — they make some arithmetic easy enough to do in your head.

(a) Expand the left side to show that $(n+1)^2 - n^2 = 2n+1$ for every number n. Then use the result to evaluate $51^2 - 50^2$ without a calculator.

(b) Use $(a + b)(a - b) = a^2 - b^2$ to evaluate 102×98 mentally. State clearly what you chose for a and for b, and show the two steps.

9. **Stretch.** Multiply each binomial by the trinomial beside it. Six products, then collect — and watch how much collapses.

(a) $(x + y)\left(x^2 - xy + y^2\right)$. State in words the pattern you notice.

(b) $(x - y)\left(x^2 + xy + y^2\right)$. How does this result compare with part (a)?

(c) Use the pattern from (a) to write the expansion of $(x + 3)\left(x^2 - 3x + 9\right)$ in one step.

Part C — Reasoning and Error Analysis

10. **Degree reasoning.** The **degree** of a polynomial in one variable is the largest exponent on the variable once the expression is fully simplified.

(a) A binomial of degree 1 is multiplied by a trinomial of degree 2. What is the degree of the product? How many products are formed before like terms are collected, and how many terms usually remain afterwards? Use $(x + 4)\left(x^2 - 3x + 5\right)$ to support your answer.

(b) What is the degree of the *sum* of that same binomial and trinomial? Explain why adding and multiplying affect the degree so differently.

(c) Is it possible for the sum of two binomials of degree 1 to have a degree *lower* than 1? Give an example or explain why not.

11. **Error analysis.** Priya simplified the expression below. Exactly **one** step contains an error.

Step 1: $4x(x+3)-(x-2)(x+5)$
Step 2: $4x^2+12x-\big(x^2+3x-10\big)$
Step 3: $4x^2+12x-x^2+3x-10$
Step 4: $3x^2+15x-10$

(a) Name the first step that is wrong. (b) Describe the error in detail. (c) Give the correct simplified expression. (d) Verify your answer by substituting $x=2$ into both the original expression and your simplified expression.

12. Simplify each expression, then **check your answer** by substituting $x=-1$ into both the original expression and your simplified version. Both must give the same value.

(a) $2x(x-3)-(x-1)(x+6)$

(b) $(3x-2)^2-9x(x-1)$

13. A very common error is to write $(x+4)^2=x^2+16$.

(a) Expand $(x+4)^2$ correctly and explain what the squaring actually does to the two terms.

(b) Find every value of x for which $(x+4)^2=x^2+16$ is actually true.

(c) By contrast, is there any value of x for which $(x+3)(x-3)=x^2-9$ is false? Explain.

Part D — Working Backwards and Solving

14. Each rectangle below has a known area and one known side. Work backwards to find the missing side, then check by multiplying.

(a) The area is $6x^2+7x-3$ and the width is $(2x+3)$. Determine the length.

(b) The area is $x^2+11x+30$ and the length is $(x+6)$. Determine the width, then write a simplified expression for the perimeter.

15. Find the missing numbers. In each case, expand the left side first and match it to the right side term by term.

(a) Find a and b so that $(x+a)(x+b) = x^2 - 2x - 15$.

(b) Find k so that $(2x + k)^2 = 4x^2 + 20x + 25$.

(c) Find m so that $(3x+m)(3x-m) = 9x^2 - 64$.

16. Solve each equation. Expand *both* sides first — in every case the x^2 terms will disappear, leaving a linear equation. One of the three has no solution; say which and explain. (If the x-terms cancel as well, there is no value of x that works: the equation has no solution.)

(a) $(x + 5)(x - 2) = (x + 1)(x + 3)$

(b) $2x(x + 3) = (2x - 1)(x + 4)$

(c) $(x + 2)(x + 6) = (x + 4)^2$

Part E — Applications

17. **Open-top box.** A rectangular sheet of cardboard measures 30 cm by 20 cm. A square of side length x cm is cut from each of the four corners, and the flaps are folded up to form an open-top box.

(a) Write an expression for the length, the width, and the height of the finished box.

(b) Write a fully expanded and simplified expression for the volume of the box.

(c) Determine the volume when $x = 4$ cm, using your expansion, and confirm it by multiplying the three dimensions directly.

18. **Path around a garden.** A rectangular garden measures $(x + 5)$ m by $(x + 3)$ m. A uniform path 2 m wide is laid all the way around the outside of the garden.

(a) Write expressions for the outer length and outer width of the garden-plus-path region.

(b) Write a simplified expression for the total area of the garden-plus-path region.

(c) Write a simplified expression for the area of the path alone. What is the degree of this expression, and why is that reasonable?

19. **Comparing two areas.** Rectangle A has length $(x + 7)$ cm and width $(x + 1)$ cm. Rectangle B is a square with side length $(x + 4)$ cm.

(a) Write a simplified expression for the area of each figure.

(b) Subtract to determine which figure has the greater area and by how much.

(c) Your answer to (b) should contain no x at all. Explain what that tells you about the two figures, and confirm it by testing $x = 1$ and $x = 10$.

20. **Volume.** Write a fully expanded expression for each volume, then evaluate it at the given value and confirm your answer by multiplying the dimensions directly.

(a) A prism has a square base of side length $(x + 1)$ cm and height $(2x - 3)$ cm. Evaluate at $x = 5$.

(b) A rectangular prism has dimensions x cm, $(x + 2)$ cm and $(x - 2)$ cm. Evaluate at $x = 6$.

Part F — Challenge

21. **Stretch: Grouping.** A trinomial in brackets can sometimes be treated as a binomial in disguise. For example, $(x + y + 2)^2$ can be written as $\big[(x + y) + 2\big]^2$, which is a perfect square whose "first term" is the whole quantity $(x + y)$.

(a) Expand $(x + y + 2)^2$ using this grouping, then finish by expanding $(x + y)^2$.

(b) Use the same idea with the difference of squares to expand $(x + y + 3)(x + y - 3)$.

22. **Three factors.** Expand $(x + 1)(x + 2)(x + 3)$ completely. Then evaluate your expansion at $x = 7$, and check it by substituting $x = 7$ into the original three factors and multiplying. Both methods must give the same number.

23. **A hidden difference of squares.** The product $\big(x^2 + x + 1\big)\big(x^2 - x + 1\big)$ looks like a trinomial times a trinomial, but regrouping each bracket as $\big(x^2 + 1\big) \pm x$ turns it into a difference of squares.

(a) Expand the product using this regrouping. You will need $\big(x^2 + 1\big)^2$ along the way.

(b) Explain why the final answer contains no x^3 term and no x term.

Answer Key. 1. (a) $3x^2-6x+x^2+3x-4-(x^2-6x+9)=3x^2+3x-13$ (b) $5x^2+15x-(2x^2+11x-6)+4x^2-4x=5x^2+15x-2x^2-11x+6+4x^2-4x=7x^2+6$ (the x-terms cancel completely). **2.** (a) $3x^2-13x-10-(x^2+8x+16)=2x^2-21x-26$ (b) $4x^2-49-(x^2+5x-24)=3x^2-5x-25$. **3.** (a) $2(x^2-9)(x+1)=(2x^2-18)(x+1)=2x^3+2x^2-18x-18$ (b) $-x(4x^2-1)=-4x^3+x$. **4.** (a) x^2+4x+4 (b) $(x+2)(x^2+4x+4)=x^3+6x^2+12x+8$ (c) $(x-1)(x^2-2x+1)=x^3-3x^2+3x-1$. **5.** $(2x^3-5x^2+5x+4)-(x^3-10x^2+25x)=x^3+5x^2-20x+4$. **6.** (a) $9x^2+12xy+4y^2$ (b) $25a^2-16b^2$ (c) x^4-9. **7.** (a) 7 (b) 4 (c) $4x$ in both blanks ($-4x$ also works). **8.** (a) $(n^2+2n+1)-n^2=2n+1$; with $n=50$, $51^2-50^2=2(50)+1=101$ (b) $a=100$, $b=2$: $102\times98=100^2-2^2=10000-4=9996$. **9.** (a) $x^3-x^2y+xy^2+x^2y-xy^2+y^3=x^3+y^3$; the four middle terms cancel in pairs, leaving a sum of cubes (b) x^3-y^3, the difference of cubes — same structure with the sign of y^3 reversed (c) x^3+27. **10.** (a) Degree 3; $2\times3=6$ products are formed, and usually 4 terms remain; $(x+4)(x^2-3x+5)=x^3+x^2-7x+20$ (b) The sum has degree 2: adding cannot create a new highest power — it can only keep the larger of the two degrees — while multiplying adds the exponents, so $1+2=3$ (c) Yes, if the x-terms cancel: $(x+3)+(-x+5)=8$, degree 0. **11.** (a) Step 3 (b) The negative in front of the bracket must be distributed to *all three* terms; Priya applied it only to x^2, leaving $+3x$ and -10 unchanged instead of $-3x$ and $+10$ (c) $4x^2+12x-x^2-3x+10=3x^2+9x+10$ (d) At $x=2$: original $=8(5)-(0)(7)=40$, and $3(4)+9(2)+10=40$. ✓ **12.** (a) $2x^2-6x-(x^2+5x-6)=x^2-11x+6$; at $x=-1$ both give 18 (b) $9x^2-12x+4-9x^2+9x=-3x+4$; at $x=-1$ both give 7. **13.** (a) $(x+4)^2=x^2+8x+16$; squaring means multiplying $(x+4)$ by itself, which produces two middle products $4x$ and $4x$ — you cannot square the terms separately (b) $x^2+8x+16=x^2+16$ gives $8x=0$, so it is true only for $x=0$ (c) No. $(x+3)(x-3)$ expands *exactly* to x^2-9, so the two sides agree for every value of x — it is an identity, not an equation to solve. **14.** (a) Length $=3x-1$, since $(2x+3)(3x-1)=6x^2+7x-3$ (b) Width $=x+5$, since $(x+6)(x+5)=x^2+11x+30$; perimeter $=2(x+6)+2(x+5)=4x+22$. **15.** (a) $a=3$ and $b=-5$ (in either order): $(x+3)(x-5)=x^2-2x-15$ (b) $k=5$ (c) $m=8$ (or $m=-8$; both give $9x^2-64$). **16.** (a) $x^2+3x-10=x^2+4x+3\Rightarrow-x=13\Rightarrow x=-13$ (b) $2x^2+6x=2x^2+7x-4\Rightarrow-x=-4\Rightarrow x=4$ (c) $x^2+8x+12=x^2+8x+16\Rightarrow12=16$, which is false — part (c) has **no solution**, since every x-term cancels as well. **17.** (a) Length $=30-2x$, width $=20-2x$, height $=x$ (b) $V=x(30-2x)(20-2x)=x(4x^2-100x+600)=4x^3-100x^2+600x$ (c) At $x=4$: $4(64)-100(16)+600(4)=256-1600+2400=1056$ cm^3; directly, $4\times22\times12=1056$ cm^3. ✓ **18.** (a) Outer length $=x+9$ m, outer width $=x+7$ m (2 m is added at *both* ends of each dimension) (b) $(x+9)(x+7)=x^2+16x+63$ m^2 (c) Path $=(x^2+16x+63)-(x^2+8x+15)=8x+48$ m^2. It has degree 1: the path is a thin border of fixed width, so its area grows in proportion to the perimeter rather than to the area. **19.** (a) A: $(x+7)(x+1)=x^2+8x+7$ cm^2; B: $(x+4)^2=x^2+8x+16$ cm^2 (b) $(x^2+8x+16)-(x^2+8x+7)=9$, so the square (B) is greater, by 9 cm^2 (c) The difference never depends on x: whatever the value, B beats A by exactly 9 cm^2. Check $x=1$: 16 vs. 25, difference 9; check $x=10$: 187 vs. 196, difference 9. ✓ **20.** (a) $V=(x+1)^2(2x-3)=(x^2+2x+1)(2x-3)=2x^3+x^2-4x-3$; at $x=5$, $250+25-20-3=252$ cm^3, and directly $6^2\times7=252$ cm^3 (b) $V=x(x+2)(x-2)=x(x^2-4)=x^3-4x$; at $x=6$, $216-24=192$ cm^3, and directly $6\times8\times4=192$ cm^3. ✓ **21.** (a) $[(x+y)+2]^2=(x+y)^2+4(x+y)+4=x^2+2xy+y^2+4x+4y+4$ (b) $[(x+y)+3][(x+y)-3]=(x+y)^2-9=x^2+2xy+y^2-9$. **22.** $(x+1)(x+2)=x^2+3x+2$, then $(x^2+3x+2)(x+3)=x^3+6x^2+11x+6$. At $x=7$: $343+294+77+6=720$; directly $8\times9\times10=720$. ✓ **23.** (a) $[(x^2+1)+x][(x^2+1)-x]=(x^2+1)^2-x^2=x^4+2x^2+1-x^2=x^4+x^2+1$ (b) The regrouping makes the product a difference of squares, and a difference of squares has no middle term at all — the two cross products $+x(x^2+1)$ and $-x(x^2+1)$ cancel exactly, and those are the only sources of x^3 and x terms.